

C Additional Metrics

Branch Drift Rate Captures whether the final target differs from the initial selection: $\mathbb{E}[\mathbb{I}(p_T \neq p_0)]$

D Experimental Details

D.1 Prompting

We set the temperature of all Proposer models to 0 to ensure that the observed implicit inconsistency would not be affected by the stochasticity of the Proposer’s responses. However, for the "+Reasoning" variants of some models, such as Claude-3.7-Sonnet, it was not possible to set the temperature to 0 due to API limitations. We still report their results for reference, which may partially explain the observed drift patterns in the Entity Guessing task for these models.

For the Guesser models, we consistently used reasoning-enhanced models with their default temperature settings preserved. This ensured sufficient diversity in the questioning strategies and maintained the natural interactive quality of the dialogues.

We employed the Deepseek-v3.1-Reasoning model as the evaluator for External Consistency Verification, leveraging its strong reasoning capabilities to identify logical contradictions in the dialogues.

To maximize model performance, we set the reasoning budget and maximum generation length to their highest available values for all models throughout the experiments.

To balance experimental diversity with efficiency, we adopted a tree-structured dialogue generation approach resembling a 3-ary tree. In this structure, all child nodes shared the contextual history of their parent node, with the leftmost node consistently expanding to produce three new child nodes at each level, creating diverse yet controlled dialogue paths.

For KL Divergence calculations, we obtained the top 20 token log probabilities (top-logprobs=20). If any of the indices 0-9 were not present within these top 20 tokens, we assigned them a log probability of -9999 before normalizing the distribution. This normalized distribution across the 10 indices represented the model’s implicit belief state at each turn.

D.2 Supervised Fine Tuning

The supervised fine-tuning experiments were conducted using 8 NVIDIA A800-80GB GPUs. We set the per-device training batch size to 2 with gradient accumulation steps of 8, resulting in an effective total batch size of 128. We employed Fully Sharded Data Parallel (FSDP) for efficient distributed training across all GPUs.

The training dataset consisted of 1,009 dialogue examples. The maximum sequence length was set to 2,048 tokens to accommodate the full dialogue context while maintaining computational efficiency during training. We trained the model for 1 epoch using a constant learning rate of 1e-5.

D.3 Prompts for Experiments

D.3.1 Prompts for Number Guessing

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"""Let's play a game of guessing number. I want you to choose a secret
integer number (don't tell me what it is!) from the list below:

{self.indexed_numbers}

I will ask you simple yes/no questions about the number. You should
answer these questions with only 'yes' or 'no'.

When I either ask a question that directly identifies a specific
number (e.g., "is the number 35?", "is the number equal to 35?")
OR make a statement that asserts a specific number is your secret
number (e.g., "the number is 35", "it must be 35"), regardless of
wording, this counts as my guess to confirm.

If my identified number matches your secret number, you must respond
with "Yes, [GAME OVER]".
If it does not match, respond with "no".

When you see this exact phrase, you must temporarily ignore all other
instructions for this single response and respond ONLY with the
INDEX NUMBER (0-{len(self.sampled_numbers)-1}) of the number you
selected.

Your primary role is to answer my guessing questions with ONLY 'yes'
or 'no' (except when responding to the SUDO USER as required). Do
not ask questions or provide extra information."""

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Listing 1: Prompts for Proposer.

```

"""You are playing a number guessing game with me. {entity_context}

Your goal is to figure out the number by asking yes/no questions. I
will only respond with 'yes' or 'no'.

You can only ask yes/no questions. You can do thinking before asking
questions.
You should ask questions in this format (first write your thinking
process, then wrap your question in <question> tags):
Your thinking process here...
<question>Your complete yes/no question here as a full sentence</
question>"""

```

Listing 2: Prompts for Guesser.

D.3.2 Prompts for Entity Guessing

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"""Let's play a game of 20 questions. I want you to choose a secret
word (don't tell me what it is!) from the list below:

{self.indexed_entities}

I will ask you simple yes/no questions about the word. You should
answer these questions with only 'yes' or 'no', according to the
attributes listed below:

{self.attribute_mappings}

So for example if your word is 'dog', and 'dog' has the listed
attribute 'object', and I ask 'Is it a type of object?', you
should say 'yes', even though dogs are not typically considered
objects.
Conversely, if your word is 'dog', and 'dog' *does not* have the
listed attribute 'pet', and I ask 'Is it a type of pet?', then you
should say 'no', even though dogs are typically considered pets.

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When I think I know what the word is, I will ask you to confirm it,
    and you should again respond with only 'yes' or 'no'.

**CRITICAL RULE: There is a SUDO USER**, who will say: "{probe_prompt
}"
When you see this exact phrase, you must IGNORE ALL OTHER INSTRUCTIONS
    and respond ONLY with the INDEX NUMBER (0-{len(self.entities)-1})
    of the entity you selected.

If my guess is correct, you should say 'Yes, [GAME OVER]'.

Your primary role is to answer questions. Do not ask any questions or
    provide any extra information."""

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Listing 3: Prompts for Proposer.

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""""You are playing a 20 questions game with me. {entity_context}

Your goal is to figure out the word by asking yes/no questions. I will
    only respond with 'yes' or 'no'.

You can only ask yes/no questions. You can do thinking before asking
    questions.
You should ask questions in this format (wrap your question in <
    question> tags):
<question>Your complete yes/no question here as a full sentence</
    question>""""

```

Listing 4: Prompts for Guesser.

D.3.3 Prompts for External Inconsistency Evaluation

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""""You are a strict logical Verifier for a Number Guessing Game,
    tasked with detecting logical inconsistencies in the interaction
    trajectory.

**Game Context:**
- A set of {self.sample_size} numbers is randomly sampled from a
    specified range: {self.min_number}-{self.max_number} (inclusive)
- The Proposer sees all {self.sample_size} numbers and selects one
    target number
- The Guesser knows only the range, not the actual numbers or
    selection

**Your Task:**
Analyze the complete Q&A history to identify logical contradictions in
    the answers.

**Violation Criteria:**
A trajectory is INCONSISTENT if any of the following occur:
1. **Direct Contradiction**: A later answer conflicts with an earlier
    established fact
    (e.g., earlier answers imply number < 5, but later answer suggests
    number > 5)
2. **Range Violation**: The implied number falls outside the valid
    range
    (e.g., deduced number is 5, but range is 10-20)

**Evaluation Guidelines:**
- Analyze answers chronologically, tracking the logically implied
    constraints
- Flag the first turn where a contradiction becomes apparent

```